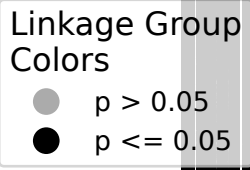


diamond: 11163 orthologs on 246 Pmax scaffolds and 4400 Rphi scaffolds.



Pmax vs Pmax groupoids

Pmax vs Rphi FET, whole chroms.

	Pmax scaf	Rphi scaf	p (FET)	Count
0	NC_047015.1	NW_026860903.1	4.05e-21	24
1	NC_047017.1	NW_026865571.1	2.83e-15	21
2	NC_047017.1	NW_026861563.1	2.09e-14	21
3	NC_047017.1	NW_026865484.1	3.23e-14	20
4	NC_047023.1	NW_026852258.1	4.64e-19	19
5	NC_047017.1	NW_026861019.1	2.47e-12	19
6	NC_047018.1	NW_026859933.1	2.50e-13	17
7	NC_047015.1	NW_026862663.1	1.99e-10	17
8	NC_047015.1	NW_026852008.1	2.20e-13	17
9	NC_047026.1	NW_026851633.1	3.97e-13	15
10	NC_047025.1	NW_026855294.1	2.51e-09	15
11	NC_047015.1	NW_026862783.1	5.23e-10	15
12	NC_047015.1	NW_026853802.1	3.53e-11	15
13	NC_047021.1	NW_026858503.1	1.45e-10	14
14	NC_047018.1	NW_026865967.1	6.86e-10	14
15	NC_047016.1	NW_026852000.1	7.55e-11	14
16	NC_047030.1	NW_026867184.1	3.47e-12	13
17	NC_047028.1	NW_026863645.1	1.14e-08	13
18	NC_047024.1	NW_026859561.1	2.99e-11	13
19	NC_047018.1	NW_026854960.1	9.49e-09	13
20	NC_047017.1	NW_026863541.1	7.44e-09	13
21	NC_047015.1	NW_026852705.1	2.52e-06	13
22	NC_047025.1	NW_026863896.1	2.86e-07	12
23	NC_047021.1	NW_026867313.1	2.82e-09	12
24	NC_047017.1	NW_026852468.1	7.16e-06	12
25	NC_047015.1	NW_026853170.1	8.60e-07	12
26	NC_047025.1	NW_026861864.1	1.88e-04	11
27	NC_047024.1	NW_026861640.1	9.86e-05	11
28	NC_047023.1	NW_026859418.1	9.07e-09	11
29	NC_047019.1	NW_026862474.1	3.63e-06	11
30	NC_047019.1	NW_026861574.1	5.22e-08	11
31	NC_047019.1	NW_026859313.1	5.22e-08	11
32	NC_047019.1	NW_026852998.1	5.22e-08	11
33	NC_047018.1	NW_026864352.1	1.78e-06	11
34	NC_047018.1	NW_026859998.1	1.58e-07	11
35	NC_047017.1	NW_026859790.1	1.26e-05	11
36	NC_047030.1	NW_026852806.1	5.28e-09	10
37	NC_047029.1	NW_026859959.1	6.71e-09	10
38	NC_047026.1	NW_026866838.1	9.41e-08	10
39	NC_047026.1	NW_026859162.1	9.41e-08	10
40	NC_047026.1	NW_026851646.1	9.41e-08	10
41	NC_047025.1	NW_026861084.1	2.24e-04	10
42	NC_047025.1	NW_026857770.1	5.47e-05	10
43	NC_047024.1	NW_026866672.1	2.03e-07	10
44	NC_047023.1	NW_026865813.1	1.04e-05	10
45	NC_047022.1	NW_026865227.1	3.76e-07	10
46	NC_047022.1	NW_026855197.1	5.07e-03	10
47	NC_047021.1	NW_026863691.1	8.01e-06	10
48	NC_047021.1	NW_026852123.1	7.70e-07	10
49	NC_047018.1	NW_026860451.1	1.36e-04	10
50	NC_047018.1	NW_026855828.1	2.42e-05	10
51	NC_047017.1	NW_026866837.1	1.43e-04	10
52	NC_047017.1	NW_026866080.1	1.40e-05	10
53	NC_047017.1	NW_026862689.1	3.18e-03	10
54	NC_047017.1	NW_026859620.1	7.93e-04	10
55	NC_047017.1	NW_026856137.1	1.40e-05	10
56	NC_047017.1	NW_026853905.1	1.40e-05	10
57	NC_047015.1	NW_026859085.1	1.13e-05	10
58	NC_047015.1	NW_026855648.1	1.16e-04	10
59	NC_047032.1	NW_026864086.1	8.42e-08	9
60	NC_047026.1	NW_026865464.1	3.69e-04	9
61	NC_047026.1	NW_026859650.1	3.69e-04	9
62	NC_047026.1	NW_026858509.1	1.83e-05	9
63	NC_047025.1	NW_026861130.1	2.79e-03	9

Pmax scaffolds

	Pmax scaf	Rphi scaf	p (FET)	Count
0	NC_047015.1 : 0-end	NW_026860903.1 : 0-end4.05e-2124		
1	NC_047017.1 : 0-end	NW_026865571.1 : 0-end2.83e-1521		
2	NC_047017.1 : 0-end	NW_026861563.1 : 0-end2.09e-1421		
3	NC_047017.1 : 0-end	NW_026865484.1 : 0-end3.23e-1420		
4	NC_047023.1 : 0-end	NW_026852258.1 : 0-end4.64e-1919		
5	NC_047017.1 : 0-end	NW_026861019.1 : 0-end2.47e-1219		
6	NC_047018.1 : 0-end	NW_026859933.1 : 0-end2.50e-1317		
7	NC_047015.1 : 0-end	NW_026862663.1 : 0-end1.99e-1017		
8	NC_047015.1 : 0-end	NW_026852008.1 : 0-end2.20e-1317		
9	NC_047026.1 : 0-end	NW_026851633.1 : 0-end3.97e-1315		
10	NC_047025.1 : 0-end	NW_02685294.1 : 0-end2.51e-0915		
11	NC_047015.1 : 0-end	NW_026862783.1 : 0-end5.23e-1015		
12	NC_047015.1 : 0-end	NW_026853802.1 : 0-end3.53e-1115		
13	NC_047021.1 : 0-end	NW_026858503.1 : 0-end1.45e-1014		
14	NC_047018.1 : 0-end	NW_026865967.1 : 0-end6.86e-1014		
15	NC_047016.1 : 0-end	NW_026852000.1 : 0-end7.55e-1114		
16	NC_047030.1 : 0-end	NW_026867184.1 : 0-end3.47e-1213		
17	NC_047028.1 : 0-end	NW_026863645.1 : 0-end1.14e-0813		
18	NC_047024.1 : 0-end	NW_026859561.1 : 0-end2.99e-1113		
19	NC_047018.1 : 0-end	NW_026854960.1 : 0-end9.49e-0913		
20	NC_047017.1 : 0-end	NW_026863541.1 : 0-end7.44e-0913		
21	NC_047015.1 : 0-end	NW_026852705.1 : 0-end2.52e-0613		
22	NC_047025.1 : 0-end	NW_026863896.1 : 0-end2.86e-0712		
23	NC_047021.1 : 0-end	NW_026867313.1 : 0-end2.82e-0912		
24	NC_047017.1 : 0-end	NW_026852468.1 : 0-end7.16e-0612		
25	NC_047015.1 : 0-end	NW_026853170.1 : 0-end8.60e-0712		
26	NC_047025.1 : 0-end	NW_026861864.1 : 0-end1.88e-0411		
27	NC_047024.1 : 0-end	NW_026861840.1 : 0-end9.86e-0511		
28	NC_047023.1 : 0-end	NW_026859418.1 : 0-end9.07e-0911		
29	NC_047019.1 : 0-end	NW_026862474.1 : 0-end3.63e-0611		
30	NC_047019.1 : 0-end	NW_026861574.1 : 0-end5.22e-0811		
31	NC_047019.1 : 0-end	NW_026859313.1 : 0-end5.22e-0811		
32	NC_047019.1 : 0-end	NW_026852998.1 : 0-end5.22e-0811		
33	NC_047018.1 : 0-end	NW_026864352.1 : 0-end1.78e-0611		
34	NC_047018.1 : 0-end	NW_02685998.1 : 0-end1.58e-0711		
35	NC_047017.1 : 0-end	NW_026859790.1 : 0-end1.26e-0511		
36	NC_047030.1 : 0-end	NW_026852806.1 : 0-end5.28e-0910		
37	NC_047029.1 : 0-end	NW_026859959.1 : 0-end6.71e-0910		
38	NC_047026.1 : 0-end	NW_026866838.1 : 0-end9.41e-0810		
39	NC_047026.1 : 0-end	NW_026859162.1 : 0-end9.41e-0810		
40	NC_047026.1 : 0-end	NW_026851646.1 : 0-end9.41e-0810		
41	NC_047025.1 : 0-end	NW_026861084.1 : 0-end2.24e-0410		
42	NC_047025.1 : 0-end	NW_026857770.1 : 0-end5.47e-0510		
43	NC_047024.1 : 0-end	NW_026866672.1 : 0-end2.03e-0710		
44	NC_047023.1 : 0-end	NW_026865813.1 : 0-end1.04e-0510		
45	NC_047022.1 : 0-end	NW_026865227.1 : 0-end3.76e-0710		
46	NC_047022.1 : 0-end	NW_026855197.1 : 0-end5.07e-0310		
47	NC_047021.1 : 0-end	NW_026863691.1 : 0-end8.01e-0610		
48	NC_047021.1 : 0-end	NW_026852123.1 : 0-end7.70e-0710		
49	NC_047018.1 : 0-end	NW_026860451.1 : 0-end1.36e-0410		
50	NC_047018.1 : 0-end	NW_026855828.1 : 0-end2.42e-0510		
51	NC_047017.1 : 0-end	NW_026866837.1 : 0-end1.43e-0410		
52	NC_047017.1 : 0-end	NW_026866080.1 : 0-end1.40e-0510		
53	NC_047017.1 : 0-end	NW_026862689.1 : 0-end3.18e-0310		
54	NC_047017.1 : 0-end	NW_026859620.1 : 0-end7.93e-0410		
55	NC_047017.1 : 0-end	NW_026856137.1 : 0-end1.40e-0510		
56	NC_047017.1 : 0-end	NW_026853905.1 : 0-end1.40e-0510		
57	NC_047015.1 : 0-end	NW_026859085.1 : 0-end1.13e-0510		
58	NC_047015.1 : 0-end	NW_026855648.1 : 0-end1.16e-0410		
59	NC_047032.1 : 0-end	NW_026864086.1 : 0-end8.42e-089		
60	NC_047026.1 : 0-end	NW_026865464.1 : 0-end3.69e-049		
61	NC_047026.1 : 0-end	NW_026859650.1 : 0-end3.69e-049		
62	NC_047026.1 : 0-end	NW_026858509.1 : 0-end1.83e-059		
63	NC_047025.1 : 0-end	NW_026861130.1 : 0-end2.79e-039		

(a) depicts the chromosomes/scaffolds of Pmax (x-axis) plotted against the chromosomes/scaffolds of Rphi (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes is the number of orthologous proteins between these two species: 11163 orthologs found between 246 Pmax scaffolds and 4400 Rphi scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (CNHs) which gene groups are most prevalent in each chromosome. (c) shows the FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group. (e) shows the FET p-value for each chromosome pair. Useful for showing if single arms of chromosomes are correlated with other regions.